

Optimal Static Native-Qubit Admission and Limits of Joint-Mode Encoding

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Abstract—We improve a previously certified static native-qubit encoder by correcting the spectrum on an unused excitation sector. With the same three qubits and interaction pairs, it reaches the same endpoint at π instead of $3\pi/2$ under a matched full centered norm-one budget. A forbidden direct-coupling bound proves this deadline optimal for the stated native input and seed. We then examine two-native-qubit admission into four single-excitation modes. An exact charge argument excludes its natural jointly charge-conserving six-qubit family. Any static two-local solution with full centered norm at most one requires time at least π for exact admission, or $\pi - 1/21845$ at complete-state error 2^{-16} . Four seeded planar numerical fits do not supply an encoder; exact rational interpretation of the best frozen fit gives an all-time error floor exceeding 0.018. A separate five-qubit odd-parity encoder reaches its different endpoint at 2π . An exact first-moment filter supplies a smaller necessary search space for the unclosed single-excitation interface.

Index Terms—Autonomous quantum control, two-local Hamiltonian, coherent admission, state transfer, spectral moment, quantum carrier.

I. OPERATIONAL TARGET AND SOURCE BOUNDARY

The preceding MQM primitive encodes one native qubit into two modes using a static planar triangle [1]. Two independent copies produce two mode excitations and do not admit a two-qubit body into the one-excitation joint carrier used by the 16-mode SWAP core. The target considered here is

$$E|ab\rangle = |ab\rangle_{AO}|\mathbf{1}_{00}\rangle_M \mapsto e^{i\phi}W|ab\rangle = e^{i\phi}|00\rangle_{AO}|\mathbf{1}_{ab}\rangle_M, \quad (1)$$

where $M = (u, v, w, z) = (m_{00}, m_{01}, m_{10}, m_{11})$. Inputs are arbitrary, including entanglement with spectators; the common phase cannot depend on ab . The six-qubit register is fully retained. We require an explicit two-local Hermitian generator, planar physical support graph, full physical norm budget, and a finite deadline window. No successful generator for this exact contract is claimed in this note.

Static gate synthesis already has spectral and numerical frameworks [2]. Transfer speed limits also have substantial prior art [3]. We derive the present scoped conditions directly rather than infer novelty from a numerical search. The norm budget concerns the full register and an explicitly chosen energy origin, not only an admitted subspace. Time is uncalibrated model time with $\hbar = 1$.

MQM development note v0.1, October 5, 2026. ORCID: 0009-0002-5813-0724. Scoped exact results and finite numerical search. Independent review and theorem-level novelty comparison pending. Physical evidence promotion: 0.

II. THE NATURAL CHARGE-PRESERVING FAMILY IS EXCLUDED

Define the bookkeeping charges

$$Q_A = N_A + N_w + N_z, \quad Q_O = N_O + N_v + N_z. \quad (2)$$

They equal the two native bits at both endpoints of (1). They are candidate conserved quantities, not axioms of MQM or physical-energy assertions.

Theorem 1 (Six-qubit charge obstruction). *Every Hermitian two-local qubit Hamiltonian on (A, O, u, v, w, z) satisfying $[H, Q_A] = [H, Q_O] = 0$ also satisfies $[H, N_z] = 0$. Consequently input 11 has complete trace error one relative to its required output for every time, regardless of coupling magnitudes or geometry.*

Proof. The per-qubit charge vectors are $(1, 0)$ for A, w , $(0, 1)$ for O, v , $(0, 0)$ for u , and $(1, 1)$ for z . A nonzero computational-basis matrix element must preserve both total charges. A two-local term changes at most two bit occupations. Flipping z changes both charges by the same sign; no single other qubit has a matching vector to cancel that change. Thus no charge-preserving two-local matrix element can change z , even after cancellation between Pauli terms. Initially $N_z = 0$; the target for input 11 has $N_z = 1$. Their supports remain orthogonal. $\square$

The replay also exhausts the complete 154-dimensional Hermitian Pauli space of support weight at most two, including imaginary matrix entries. An integer commutator Gram matrix has a 38-dimensional charge-preserving kernel. All 38 exact kernel vectors commute with N_z . This is a family exclusion, not a universal impossibility theorem for two-local admission. Breaking at least one stated charge, changing the carrier, admitting higher-body support, or adding counted resources changes the premises.

III. A DEADLINE BOUND WITHOUT CONSERVED-CHARGE ASSUMPTIONS

For nonzero native inputs, source and target bit strings in (1) differ in three occupations (01, 10) or four (11). Thus every two-local generator has zero direct source–target matrix element. The following lemma applies to arbitrary Hermitian matrices, including complex ones.

Theorem 2 (Forbidden direct-coupling deadline). *Let normalized states e, f obey $\langle e, f \rangle = 0$ and $\langle e, Hf \rangle = 0$, and let*

$\|H\| \leq 1$. If the pure-state trace distance from $e^{-itH}e$ to f is at most $0 \leq \epsilon < 1$, then

$$t \geq L(\epsilon) := \pi - 2 \arctan\left(\frac{r_\epsilon}{1-\epsilon}\right), \quad (3)$$

where $r_\epsilon = \sqrt{2 - 2\sqrt{1-\epsilon^2}}$. In particular, exact transfer requires $t \geq \pi$.

Proof. For $t \geq \pi$ the bound is immediate. Assume $0 < t < \pi$ and write $a = e^{-itH}e$. Choose a phase ζ with $\|a - \zeta f\| \leq r_\epsilon$. Orthogonality gives $|\langle e, a \rangle| \leq \epsilon$, and the zero direct matrix element gives

$$|\langle e, Ha \rangle| \leq \|He\| r_\epsilon \leq r_\epsilon. \quad (4)$$

For $\lambda \in [-1, 1]$, scalar monotonicity yields

$$\lambda \sin(t\lambda) \geq \cot(t/2) [1 - \cos(t\lambda)]. \quad (5)$$

Indeed $y \cot(ty/2)$ decreases for $0 < y \leq 1$; its derivative sign follows from $x - \sin x \cos x \geq 0$, whose derivative is $2 \sin^2 x$. By spectral calculus,

$$\begin{aligned} r_\epsilon &\geq \langle e, H \sin(tH)e \rangle \\ &\geq \cot(t/2) [1 - \operatorname{Re}\langle e, a \rangle] \geq (1 - \epsilon) \cot(t/2). \end{aligned} \quad (6)$$

Rearranging gives (3). At $\epsilon = 0$, the positive spectral function would force $He = 0$ for $t < \pi$, contradicting orthogonal transfer. $\square$

An arbitrary energy shift leaves both zero matrix elements and the channel unchanged. Hence the result applies whenever $\|H - xI\| \leq 1$ for a declared origin x . It uses neither graph planarity nor the charges of Section II.

The full admission half-diamond error dominates its trace error on input 11. Consequently any static two-local solution to (1) obeys the same lower bound. At $\epsilon = 2^{-16}$, $r_\epsilon \leq \sqrt{2}\epsilon < 3\epsilon/2$ and $\arctan y \leq y$ give the exact conservative form

$$T \geq \pi - \frac{3\epsilon}{1-\epsilon} = \pi - \frac{1}{21845}, \quad 21845 = 5 \cdot 17 \cdot 257. \quad (7)$$

This is a necessary bound, not an achieved six-qubit optimum. The scalar lemma's exact π limit is sharp: a three-level path with edge weights $1/\sqrt{2}$ has norm one, no direct endpoint coupling, and transfers endpoints at π . Its sharpness does not establish saturation of the complete four-input admission contract. The proof assumes a single time-independent generator; it does not certify time-dependent schedules or dissipative models.

IV. A MATCHED OPTIMAL IMPROVEMENT OF ONE-QUBIT ADMISSION

The deadline bound applies to the preceding one-qubit task as well: its source $|110\rangle_{Auv}$ and target $|001\rangle_{Auv}$ differ on three qubits. No two-local generator has a direct matrix element between them. Thus exact static admission under the full centered norm-one budget cannot finish before π .

Retain the parent interaction triangle [1]. With $S = N_A + N_v$, its uncentered generator is $K_g = gX_u S + (g/2)(X_A X_v + Y_A Y_v) + 2gS$. Add a diagonal correction on an existing pair:

$$H_g^* = K_g - 2gN_A N_v - 2gI. \quad (8)$$

The added density term vanishes throughout the admitted $N_A + N_v \leq 1$ sectors. Their uncentered evolution is unchanged. It shifts the unused double-occupation sector down by $2g$, reducing the full spectral width from $6g$ to $4g$ without adding qubits or pairs.

Theorem 3 (Optimal isolated native-qubit admission). At $g = 1/2$, H_g^* is a planar three-qubit two-local generator with full norm one and spectrum $(-1, 0, 1)$ of multiplicities $(4, 2, 2)$. At $T = \pi$,

$$e^{-i\pi H_g^*} |\psi\rangle_A |10\rangle_{uv} = -|0\rangle_A (\alpha|10\rangle + \beta|01\rangle)_{uv}. \quad (9)$$

The same duration decodes. The deadline is optimal among static Hermitian two-local generators for this specified input/seed contract with full centered norm at most one.

Proof. In the admitted zero-charge sector the centered generator is $-2gI$. In its one-charge sector it is $g(X_u + X_{Av})$. At $gT = \pi/2$, both cases produce the stated common minus sign. The double-charge sector is $2gX_u$, also bounded by $2g$. The full characteristic polynomial of H_g^*/g is $(z+2)^4 z^2 (z-2)^2$, proving the claimed spectrum. The earlier deadline theorem supplies $T \geq \pi$, and this construction attains it. The same isometry argument includes arbitrary spectrators. $\square$

All Pauli coefficients at $g = 1/2$ are dyadic, so exact binary64 coefficient implementation introduces no rounding debit. The readout interval $|t - \pi| \leq 2^{-18}$ has complete error at most $2^{-18} < 2^{-16}$ under ideal admission and implementation. Recurrence and readout timing remain; no permanent latch is supplied.

The latency ratio is exactly $\pi/(3\pi/2) = 2/3$, a reduction by one third under the matched global norm budget. The graph stays three vertices, three pairs, degree two; the Pauli count increases from eight to nine. The largest Pauli coefficient increases from $1/3$ to $1/2$ (largest two-body coefficient from $1/6$ to $1/4$). At fixed underlying $g = 1/3$, the correction instead lowers full norm from one to $2/3$ with the original deadline unchanged. Therefore this is not a speed improvement under a separately fixed per-term ceiling. It is a model-level resource certificate, with independent novelty comparison and physical realization pending.

V. SPECTRAL MOMENTS AND A FINITE PLANAR SEARCH

If $e^{-iTH}E = e^{i\phi}W$, then for every nonnegative integer k ,

$$E^\dagger H^k E = W^\dagger H^k W. \quad (10)$$

This follows because the propagator commutes with H . The first moment is a linear constraint on Pauli coefficients. Its exact rank is 15 in the full 154-dimensional two-local space, leaving dimension 139. Higher moments and eigenphase alignment remain necessary; first-moment matching is not sufficient.

Our exploratory fit uses 62 real Pauli coefficients: X and Z fields and XX, YY, ZZ, XZ, ZX on a fixed planar ten-pair graph. In order (A, O, u, v, w, z) its pairs are

$$\{Au, Aw, Az, Ou, Ov, Ow, Oz, uw, uv, wz\}. \quad (11)$$

TABLE I
FINITE NUMERICAL SEARCH DIAGNOSTICS; NO EXACT ENCODER

Start	Evaluations	Isometry norm residual
0	424	0.783433
1	1000	0.577244
2	483	0.533405
3	570	0.704262

Its maximum degree is four. This ansatz permits charge breaking but omits other allowed complex couplings. We minimize the full four-column isometry residual with one common target phase. Four starts use seed 20261005, coefficient coordinates in $[-4\pi, 4\pi]$, and a cap of 1000 function evaluations per start. An analytic exponential derivative is checked against finite differences. The coordinate box is not a physical norm budget; fitted generators are normalized separately.

The best fitted endpoint residual is 0.533405. These local fits do not exhaust the parameter box, all planar graphs, or all two-local generators. Three stopped under the loss tolerance; one reached the evaluation cap. None certifies the requested 2^{-16} target.

A. Exact all-time rejection of the frozen best fit

We interpret its binary64 coefficients c_P as exact binary rationals and set $B = \sum_P |c_P|$, so $\|H\| \leq B$. For input 11, let $\Delta = \langle e, He \rangle - \langle f, Hf \rangle$. Static evolution conserves the first expectation. Trace duality therefore gives, for every time,

$$D(e^{-itH}e, f) \geq \frac{|\Delta|}{2B} > \frac{18}{1000} > 2^{-16}. \quad (12)$$

The exact fraction is in the receipt; its decimal value is 0.018247196. This rejects that one frozen generator at all deadlines, including after uniform norm rescaling. It is not a lower bound for other coefficients in the search ansatz.

For the next attempt we supply an exact kernel of the first-moment constraints inside the same 62-parameter ansatz. Its rank is seven, leaving a 55-dimensional parametrization. This prevents the next fit from spending effort on generators already excluded by first moments, while leaving higher moments and endpoint phases to be solved.

VI. AN EXACT ALTERNATIVE CARRIER, NOT THE REQUESTED ENCODER

Changing the output carrier avoids the charge confluence. Use five qubits (A, O, u, v, w) , define $S_A = N_A + N_w$, $S_O = N_O + N_v$, and set

$$\begin{aligned} \tilde{H}_g = & gX_u(S_A + S_O) + 2g(S_A + S_O) \\ & + \frac{g}{2}(X_AX_w + Y_AY_w + X_OX_v + Y_OY_v) \\ & - 2g(N_A N_w + N_O N_v) - 4gI. \end{aligned} \quad (13)$$

The two native transport pairs and the shared seed terms commute as grouped operators. Its physical graph is two triangles sharing u : five qubits, six pairs, maximum degree four, planar. All terms are at most two-local.

Proposition 1 (Odd-parity endpoint admission). *At $g = 1/4$ this full-register generator has norm one and width two. At $T = 2\pi$ it maps*

$$|ab\rangle_{AO}|100\rangle_{uvw} \mapsto |00\rangle_{AO}|1 \oplus a \oplus b, b, a\rangle_{uvw}. \quad (14)$$

The same duration decodes, uniformly over spectators.

Proof. For admitted source bits, $S_A = a, S_O = b$ and the density correction vanishes. At $gT = \pi/2$, transport moves the native occupations into w, v , while the seed flips by the parity of $a+b$. The $2g(S_A + S_O)$ phases cancel the transport and seed phases, and the origin shift supplies phase one. Thus the output mode words are 100, 010, 001, 111. Direct full characteristic-polynomial verification gives eigenvalues of $\tilde{H}_g/g$ equal to

$$(-4, -2, 0, 2, 4) \quad (15)$$

with multiplicities (10, 8, 6, 4, 4). Their full norm and width at $g = 1/4$ are one and two. All endpoint eigenphases are ± 1 , giving a full return at $2T$ and the same-generator decoder. $\square$

All Pauli coefficients at $g = 1/4$ are dyadic. The nominal window $|t - 2\pi| \leq 2^{-18}$ has complete error at most $2^{-18} < 2^{-16}$ under ideal admission and implementation.

This odd-parity carrier has one or three excitations at the endpoint. Its parity is not asserted invariant during admission. It cannot be connected directly to the existing single-excitation SWAP core, and no static admission–execution–extraction episode is established. A comparison with that core must include the conversion and any new computation encoding. Bare-energy realization, coherent supply, irreversible retention, and clock refresh remain open.

VII. VERIFICATION AND NEXT GATE

The replay records 78 exact checks and 4 numerical propagation diagnostics, separately from the four exploratory fits. Exact checks cover the entire two-local charge kernel, support-weight zero couplings, scalar proof identities, sharpness example, full parity spectrum, spectator isometry, rounding, moment-filter kernel, and frozen energy witness. Independent review and theorem-level novelty comparison are pending.

The next target remains the six-qubit single-excitation map (1). Use the exact moment filter, test higher moments and coherent endpoint phases, and broaden the physical ansatz only with its graph and norm ledger explicit. A failed numerical fit is not an impossibility result. The five-qubit alternative is a separate exact comparator, not a promotion of the six-qubit contract or a demonstrated computational advantage.

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